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Open-source CSIP-AUS to OCPP demonstration

Scoping note for the OCA Plugfest Australia, 21–23 October 2026

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Purpose Scope a public, reproducible demonstration that exercises all three protocols in draft AS 5438. SEP2 and OCPP 2.1 are the primary paths, SunSpec Modbus is the secondary path. All components are open source and already running in the field.

Draft AS 5438 asks an inverter energy management system to work across three protocols end to end. Once the standard is finalised, an EEMS will need to accept a CSIP-AUS envelope from a distribution network operator, coordinate the household through SunSpec Modbus, and drive a bidirectional charger through OCPP 2.1. Several Australian trials cover parts of this path, but a fully open-source reference that runs the whole chain, including the OCPP 2.1 bidirectional leg, would be a useful addition to the evidence base.

This note proposes such a demonstration at the Plugfest. All three AS 5438 protocols are exercised on a single household stack: SEP2 (via CSIP-AUS) and OCPP 2.1 as the primary paths, SunSpec Modbus as the secondary path. Every component is open source, and the stack is running today on a Sunshine Coast site in Queensland.

What this note asks

Confirm the Plugfest can host a CSIP-AUS-to-OCPP demonstration using open-source components.

Agree the AS 5438 clauses the demonstration should exercise, so it produces evidence useful to the standard.

Nominate a 30-minute slot in the Plugfest programme for a live walk-through and questions.

Note that the HAEO and ODE maintainers have been consulted on the technical approach but are not confirmed contributors to the Plugfest scope.

Why this matters now

Draft AS 5438 *Interoperability Requirements for Inverter Energy Systems* closed public comment on 22 April 2026. The draft names three protocols for inverters and any coordinating platform:

- SunSpec Modbus 1.2, over Ethernet or RS-485.
- SEP2, adopted in Australia as AS 5385:2023 and based on IEEE 2030.5:2018, over Ethernet.
- OCPP 2.1, over Ethernet.

Inverters must support at least one. A platform that coordinates multiple inverters, an electrical energy management system or EEMS, must support all three.

OCPP 2.1 is the AS 5438 protocol for bidirectional EV chargers. Most bidirectional field trials to date have used vendor protocols or DNP3. AS 5438 asks the industry to prove that the open OCPP path works end to end, from a network operator envelope down to a charging profile at the socket. The Plugfest is the natural venue to gather that evidence.

The demonstration stack

All components are open source. All are running on a Sunshine Coast site in Queensland today. The coordinating platform is a Home Assistant Green appliance running the ODE add-on and the HAEO optimiser (Figure 1).

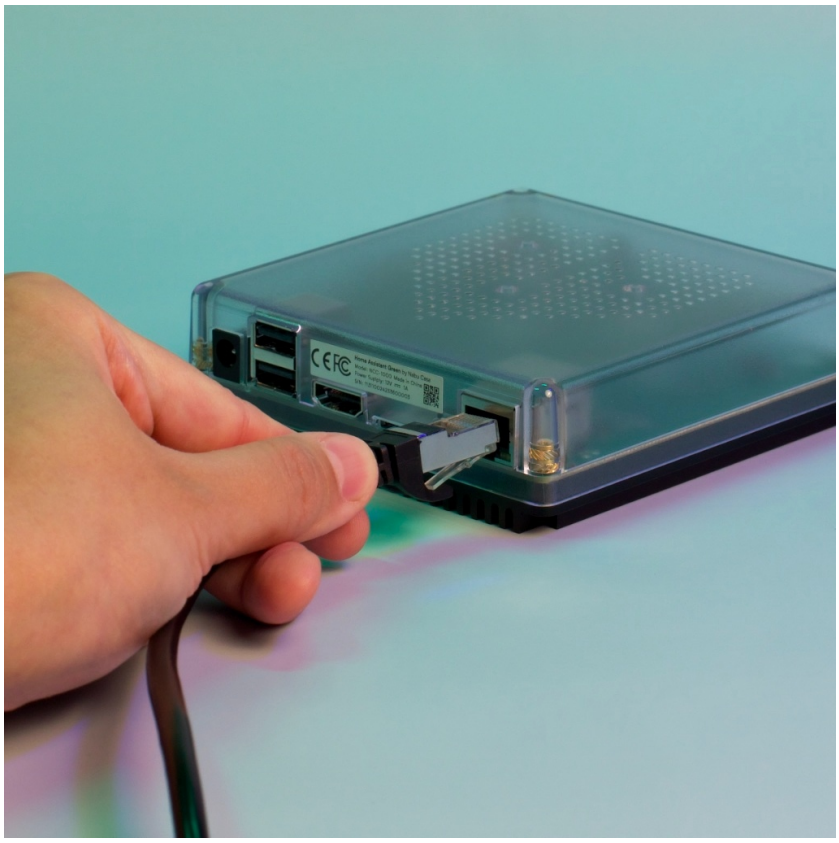


Figure 1. Home Assistant Green. The coordinating platform runs on a Home Assistant Green appliance on the Sunshine Coast site. The ODE add-on and HAEO optimiser both run inside Home Assistant on this device.

Layer	Component	Role in the demonstration
Network operator server	CSIP-AUS / SEP2 server, or ANU BSGIP CACTUS test harness	Publishes the envelope the household must respect.
CSIP-AUS client	Open Dynamic Export (ODE)	Certified CSIP-AUS client. Turns the envelope into MQTT and REST feeds.
Coordinating platform (EEMS)	Home Assistant Green with ODE add-on	Speaks all three AS 5438 protocols. Ingests the envelope, calls the optimiser, sends setpoints out.
Optimiser	HAEO (Home Assistant Energy Optimizer)	Solves a linear program over prices, forecasts and the envelope. Produces a 24-hour household plan.
Price signal	AEMO NEMWEB spot price feed	AEMO Queensland spot price is used as the wholesale price forecast that HAEO optimises against.
SunSpec client	Home Assistant SunSpec integration	Reads and writes SunSpec Modbus 1.2 to the hybrid inverter.
OCPP client	lbbrhzn/ocpp Home Assistant integration	Sends the plan to the charger as an OCPP TxProfile.
Hardware	Hybrid inverter, battery, PV, bidirectional OCPP charger, EV	The physical energy resources on site.

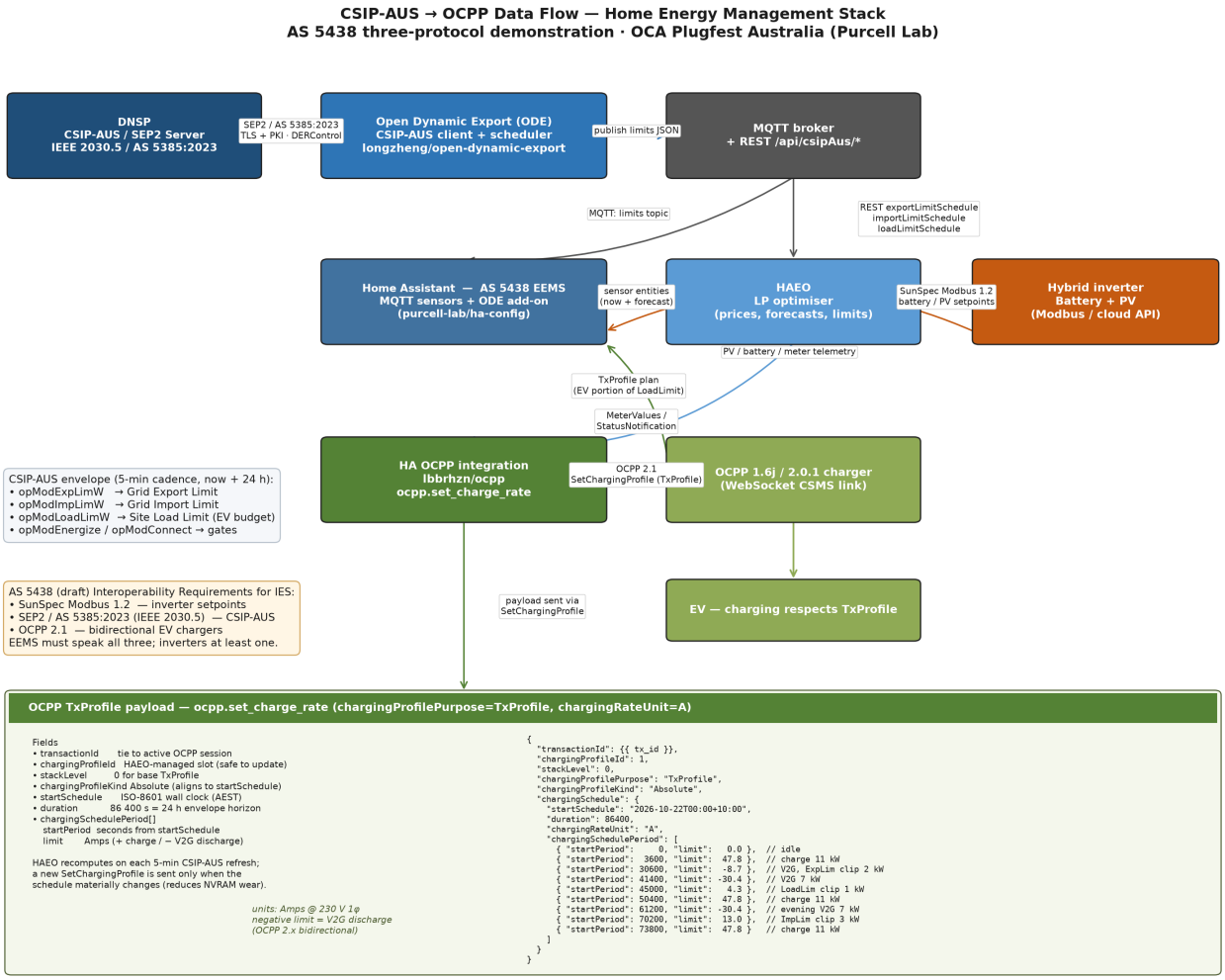


Figure 2. Data flow across the stack. The three AS 5438 protocols are labelled on the paths that carry them. Home Assistant plays the AS 5438 EEMS role. It is the only component that speaks all three.

How the demonstration maps to AS 5438

AS 5438 protocol	Physical layer	Where it appears	In this demo?
SunSpec Modbus 1.2	Ethernet or RS-485	HA SunSpec integration to hybrid inverter	Yes, secondary path
SEP2 (AS 5385:2023, IEEE 2030.5)	Ethernet	Network operator to ODE (CSIP-AUS profile)	Yes, primary path
OCPP 2.1	Ethernet	HA OCPP integration to bidirectional charger	Yes, primary path

All three protocols are exercised. SEP2 (via CSIP-AUS) and OCPP 2.1 are the primary paths. They carry the network envelope and the bidirectional charging control that draft AS 5438 most needs to evidence. SunSpec Modbus 1.2 is the secondary path, exercised through the Home Assistant SunSpec integration against the hybrid inverter.

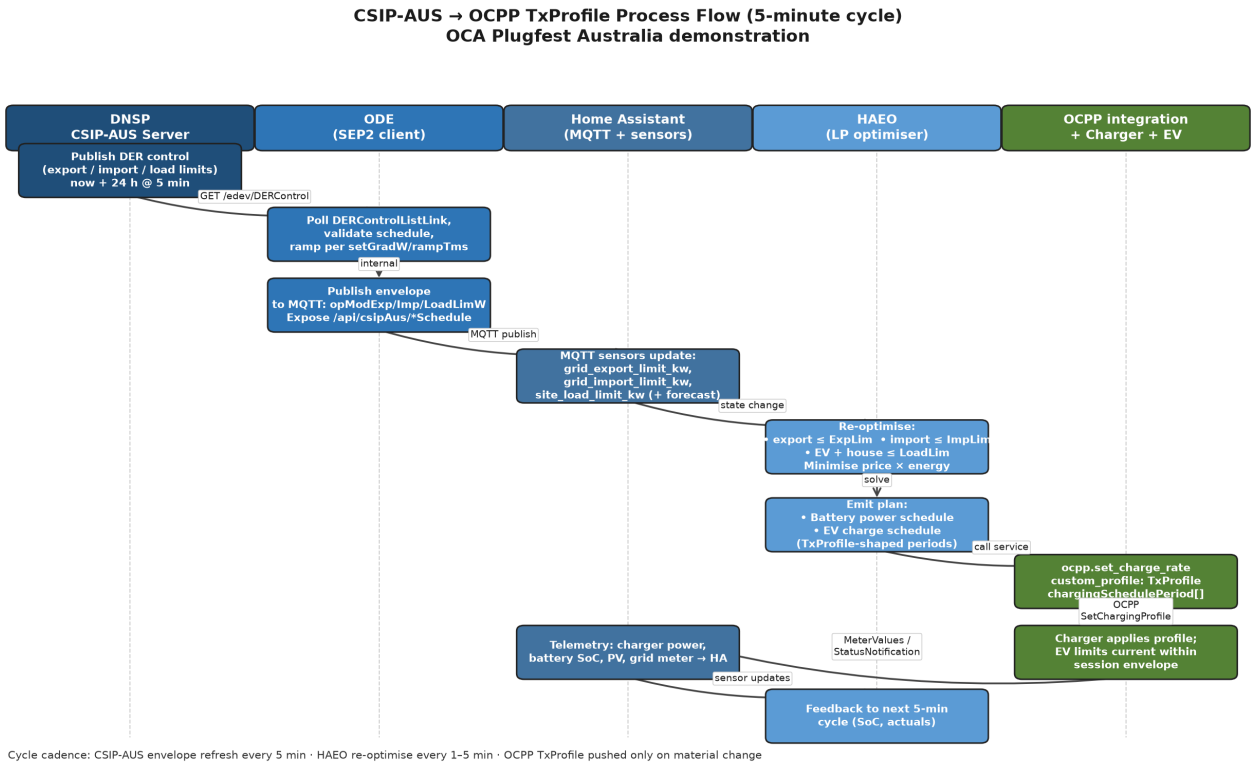


Figure 3. The five-minute control cycle. Every five minutes the network operator issues a fresh CSIP-AUS envelope. ODE republishes it. HAEO re-solves the 24-hour plan. Only if the schedule has materially changed does the OCPP integration send a new SetChargingProfile to the charger. This cadence protects the charger's non-volatile memory.

What the envelope looks like in practice

Figure 3 traces one day of operation. The top panel shows the CSIP-AUS envelope (export limit, import limit and load limit), with V2G discharge plotted as a negative export flow. The middle panel is the household plan. The bottom panel is the OCPP TxProfile the charger receives, expressed in amps at 230 V single-phase.

Three constraint scenarios are shown. An export-limit clip during the morning V2G discharge. An import-limit clip during afternoon charging. A load-limit clip at the household level. Each scenario shows the clip and the return to the normal setpoint. This is the behaviour AS 5438 clause tests will need to see.

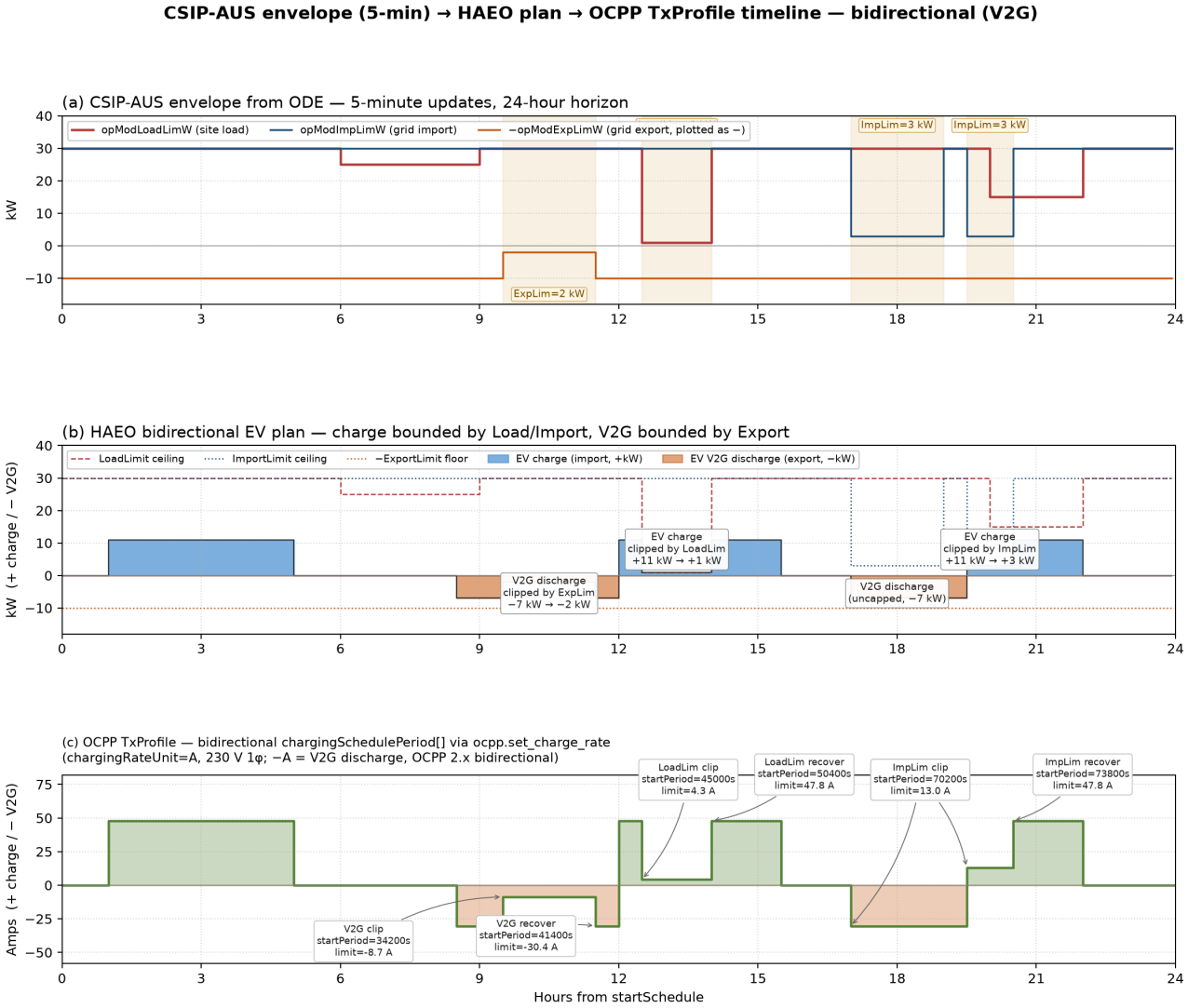


Figure 4. Bidirectional worked example. Envelope, household plan and OCPP TxProfile shown on a common time axis for 24 hours. V2G discharge, grid import and household load are each constrained on a separate CSIP-AUS channel, and each constraint is visible in the TxProfile the charger receives.

What the Plugfest gets from this

The Plugfest is about interoperability evidence. This demonstration produces four kinds of evidence in a single 30-minute session:

- **End-to-end trace.** A CSIP-AUS DERControl event on the wire, the MQTT and REST hop through ODE, the household plan, and the resulting OCPP SetChargingProfile, all timestamped and reproducible.
- **Bidirectional coverage.** V2G discharge under an export limit is shown, not just one-way charging under an import limit. The AS 5438 choice of OCPP 2.1 is largely about bidirectional support.
- **Timing evidence.** Read and write response times, active-power limit achievement, and communication-loss fail-safe behaviour can be measured against the draft AS 5438 function map (≤5 s, ≤30 s, ≤15 s respectively).

• **Open reference implementation.** Every component is public. Any vendor at the Plugfest can point a client at the same test harness and compare results.

What is in scope, and what is not

In scope	Out of scope
CSIP-AUS envelope ingestion through ODE. Household plan generation by HAEO. OCPP 2.1 SetChargingProfile delivery. Bidirectional operation (V2G discharge and EV charging). Three constraint scenarios: export limit, import limit, load limit. Timing measurements against the AS 5438 draft function map.	Vendor certification of the hybrid inverter. Formal conformance testing under IEEE 1547.1. Long-run reliability data. This is a Plugfest demonstration, not a field trial. Confirmed contribution from the HAEO or ODE maintainers. They have been consulted, not scoped in.

Known gaps to flag on the day

• ODE has two open CACTUS test gaps that would be named up front: LOA-05 (IEEE 2030.5 Subscription support) and GEN-11 (EventSuperseded handling for interrupted controls).

• The three-protocol coordination has not been independently tested against the AS 5438 function map. The Plugfest is the natural venue to start that evidence base.

• OCPP 2.1 is still experimental in the lbbrhzn/ocpp integration. The demonstration will call this out. A fall-back to OCPP 2.0.1 can be shown if a charger on the day requires it, but only for one-way charging. The bidirectional (V2X) charging profile is an OCPP 2.1 feature and does not exist in 2.0.1 or 1.6.

Consultation with upstream maintainers

Two upstream maintainer teams have been consulted on the technical approach. They are **not** confirmed contributors to the Plugfest scope; their involvement on the day would need separate agreement.

• **HAEO maintainers**, **Trent Houliston** and **Brendan Annabel**. Consulted on the linear-program formulation, the five-minute re-solve cadence, and the timing instrumentation needed to measure the AS 5438 function map (≤5 s read/write, ≤30 s active-power limit, ≤15 s communication-loss fail-safe).

• **ODE maintainer**, **Long Zheng** (longzheng/open-dynamic-export). Consulted on CSIP-AUS envelope publication, the MQTT and REST feed shapes, and the two open CACTUS test gaps that would be named on the day.

This consultation shapes the reference path the demonstration will follow. It does not commit any of the maintainers to Plugfest work, and no action is required from the Plugfest organisers in relation to these teams.

Suggested next steps

- Confirm whether the Plugfest programme can host a 30-minute open-source demonstration slot.
- Agree the AS 5438 clauses the demonstration should exercise, so timing measurements produce evidence that is directly useful to the responsible Standards Australia committee.
- Confirm the OCPP versions the Plugfest will accept for a bidirectional demonstration, and whether the test harness can log a full SetChargingProfile trace for later publication.

A 30-minute call in the fortnight starting 3 August 2026 would be enough to walk through the scope in detail. The full stack can also be shown over a screen share, so the envelope-to-TxProfile path can be seen end to end before deciding.

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References

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- ANU BSGIP CACTUS, CSIP-AUS communications client test procedures. bsgip.com